All right. Good afternoon. Just a reminder, quiz one is due today. So you can do that on canvas if you haven't yet. You have up to an hour from when you start it. It probably won't take the full time. Yeah. It does. Yeah. So it's due today. But if for some reason you don't submit it today, it will be accepted still until end of day tomorrow. Just a quick recap of last time. So we talked about the difference in frequentist and basing inference, two different types of likelihood-based inference. We started with just some important definitions, the distinction between a population and a sample, right? A population is kind of fixed, but there's many quantities. If we could observe them all, we wouldn't need statistics, but instead we only get to observe a sample, a small subset of the observations, right, and that sample is random. And so that randomness is where statistics and probability comes in. Parameter is some numerical measurement associated with the population. Usually the thing we're interested in estimating and learning about the population, but it could also just be a parameter of the distribution that we use for our model. Visualized this distinction between the sample and a population, and then we talked about the two different types of inference. So frequentist inference, we're treating our parameter as some fixed but unknown quantity. There's no randomness. We don't think of parameters as random when we have our frequentist hat on. There's just some fixed number that we don't know. we'd like to learn about. Bayesian's actually treat the parameters as random. Whether or not they actually believe they're random isn't really the point. We treat them as random. We're willing to put a probability distribution on these. Specifically, we have a distribution before we observe any data. We call that the prior distribution. And what we'll talk about today is how we update our beliefs. After we've observed some data, we get what's called the posterior distribution. But the point is we're treating these parameters as random variables, right? And so they have a probability distribution. Very different from the frequentist mindset where there's no randomness associated with the parameter. It's just some fixed but unknown quantity. So we'll have to transition back and forth between those two mindsets throughout the course. So just keep that in mind, fixed versus random, frequentist, Basian. They are both examples of likelihood-based inference. They both use the likelihood, albeit in different ways. So we gave a definition for the likelihood, specifically for the case where we had a random sample, our observations are independent observations from the same distribution, from the same population, right? And this is the likelihood function. It's just a product of these probability density or probability mass functions, right? But we're now viewing it as a function of theta. We're kind of thinking of x, the data is fixed. We've observed the data, right? And we're varying theta, varying being the keyword there. Whether or not it's random, it doesn't matter we can vary theta. It's a function of theta. We saw that these indicator functions can be very useful when writing down a likelihood, right? It can identify parameter restrictions, the parameter or space. And so this like or this indicator function i just returns one if the thing in the parentheses is true, zero otherwise.

So we looked at an example of this likelihood. Just a reminder, a likelihood is not a probability distribution. We're not saying anything about probability of the parameter p here. It's just some relative notion of how likely one value is compared to another.

These values here are relatively likely. These values are relatively unlikely. Yeah, question.

A loss function is more general. A likelihood can be a loss function. It's a specific type of loss function. But yeah, there's other types of loss functions that aren't likelihoods.

Yeah, good question. You can. That's another framework called fiducial inference. Not really what we're talking about here. It's not used so much, but it does exist. Yeah, good question. And then actually I'll show a plot today where I do kind of scale the likelihood just to make it more compatible with the prior and the posterior when we look at that as well. Looked at some examples of just writing down a likelihood question. It's all relative. Yeah, the magnitude

in itself doesn't really have any real meaning. It's just all relative. And then we kind of, so we said, you know, both these frameworks, frequentist and inference are using the likelihood. But in this course, we're going to start with Bayesian inference, and so we kind of transition, and we talked about the prior distribution. This is the distribution we put on our parameter. The parameter is random. The distribution we put on our parameter before we've observed any data. That's the prior distribution, right? It could be a probability density function or a probability mass function, depending on whether that parameter is continuous or discrete. And we saw some important examples here. In this first example, we need to consider the parameter space, what are the possible values the parameter can take. In this coin flip example, Theta, the probability of heads could only be 0.5 or 1, our prior should reflect that. It should only put positive probability mass on the values 0.5 or 1, and it should return 0 otherwise. If we know those are the only two possible values, we shouldn't give positive probability or positive probability density in the continuous case for values that aren't allowed. We talked about a case where we didn't really have any prior information to go off of right for this parameter theta that was the proportion of defective iPhones right we just knew it had to be between zero and one because it's proportion so we used a uniform prior here we're not giving any favoritism to any single value or range of values right within that interval just uniform flat prior and then we looked at another example where maybe we did have some prior information we'd like to include that right maybe we thought before observing any data we thought the mean, a good guess for the mean is 0.002, and we're fairly confident in that, right? So we have this standard deviation that's pretty small, 0.001, right? And we wanted a gamma prior distribution, but the gamma prior distribution is parameterized in terms of alpha and beta, not the mean and standard deviation.

Right. So we solved this system of two equations to get those parameters alpha and beta associated with our prior distribution. Okay, so we talked a lot about constructing an appropriate prior. We didn't talk at all about what you do after that, how we actually do Bayesian inference, right, after we've observed the data. And so that's going to be the topic today, the posterior distribution.

It's, as we just said, the prior distribution, which we denote as pi of theta, is our state of knowledge, everything we believe about the parameter theta before we observe any of the data, right? But this is statistics. We're going to observe a sample of data. We want to use that sample to infer something about the population or the population parameters, right? And so we're going to work with the posterior distribution. Okay, so let's define that. So consider a statistical inference problem. A parameter of interest is theta. Could be a vector. It doesn't have to be, right? And we have random variables that we're going to observe X1 through XN.

Then the conditional distribution of theta, given the data X, and we can use a conditional distribution, again, because theta is random with our basian hats on.

This conditional distribution is called the posterior distribution, and then the conditional PDF or probability mass function of theta, given specific values, so the first random variable equal to this specific value, little X1, so on through the last random variable, big XN, equal to some specific value, little XN, the actual values of the data we observe. That's the posterior PDF or posterior PMF of theta denoted in this way. It's just a conditional distribution like you learned about in probability theory, right?

Only now the distribution is over a parameter. It's a little different than probability theory. So this posterior distribution is going to, similar to the prior, it's going to express our state of knowledge, but it's going to do so after we've observed the data. We're updating our state of knowledge. We're updating our beliefs about the parameter. But it is a probability distribution. It's a conditional distribution, right? So we can do all the things that we do with probability distributions. We could calculate the mean of the distribution. We could calculate the variance of the distribution. Anything else that you do

with the distribution, we can do because we have a distribution for the parameter.

So if we think about high posterior variance, meaning this posterior distribution has a very large variance, right? That just means we're not very certain about what values are likely. There's a lot of uncertainty still after we've observe the data. On the flip side, if we have low posterior variance or small posterior variance, we don't have a lot of uncertainty. We feel fairly confident about our state of knowledge for the parameter.

And that's all expressed in this posterior distribution. And so Bayesian inference is really centered around this posterior distribution. The main goal is to get this posterior distribution. And how do we get there? Well, base theorem. We learned this in probability theory. Again, we weren't doing it with a distribution of our parameter, but it's the same theorem, right?

It's just a conditional probability distribution. We want the conditional distribution of theta given the data X. And that's just base theorem. So this tells us that this posterior, pi of theta given X1 through XN, the data is equal to the product of our distributions for the data given the parameter, X1 given the parameter through XN given the parameter, right? If we have a random sample times our prior distribution, what we called pi of theta, and then divided by this function of the data, M.

And notice the parameter doesn't show up here in the denominator. So we call this the joint marginal distribution of the data, right? It's marginalized out. There's no theta involved, no parameter involved. And so if we're thinking of a distribution over theta, right, this is constant with respect to theta. So that's important. I'll talk about that A little more in a second, yeah. Because we've integrated out the parameter, integrated out theta, right? But it's the joint marginal because it's still joint across all X1 through XN. So recall for probability theory, you can marginalize out a variable, get a marginal distribution from a joint distribution. Here we have a joint distribution of all the data and the parameter. We can marginalize out that parameter, integrate out the parameter if it's continuous. Either the prior or the posterior, but when they say, like, you collected this. amount of, like, let's see you like this amount of data that follows another type of distribution. That just refers to the likelihood, right? Exactly. When we talk about the distribution of the data given the parameter, that's the likelihood, right? And then if we're talking about just the distribution of the parameter itself before observing any data, that's the prior, after observing the data, that's this posterior, that we have based theorem to calcium. So different distributions floating around, you know, keep track of which is which. But, yeah, exactly. The distribution of the data, given the parameter, F of x1 given theta is the likelihood. Well, collectively this makes up the likelihood. And so this is for a random sample specifically. Again, independent draws all from the same population or the same distribution. And so when we write down the joint distribution of our data given the parameter, because of independence, because it's a random sample, we can write this as a product, product of these individual density or mass function. One notation thing, so here we have x1 through xN. I'll often write this as a vector, collectively as a vector X, and I will bold when I'm using a vector. Bold is a vector. Unbolded, we see the individual scalar terms that make up the vector. But again, this is a random sample, which is why this joint distribution conditional in the parameter is a product of these marginals. And then as I just said, this denominator in Bayes theorem, this M, right, this is the marginal distribution of the data, right, we'd get it by marginalizing or integrating out the parameter, right? And the parameter, pi of theta, is our prior distribution, and we can integrate out over the parameter space for theta, the allowable values of theta. And we call this M, or again, we can use bold X to denote the vector of X's here. And so again, notice, after we've integrated out parameter, it's no longer a function of theta, right? There's no theta involved here. We've integrated it out. It's just a constant in terms of theta. So it's a constant term. It shows up in the posterior distribution, basically to enforce the fact that the probability distribution integrates or sums to one. We're not usually

interested in this, though. It's, if anything, it's usually more of a nuisance. In a lot of the examples today, we won't really even need to look at it, and I'll show you why. So since we ignore it, since it's a constant term, right, instead of writing what the posterior distribution is explicitly equal to, we will very often use this instead. And so we have this symbol here. This just means is proportional to. Is proportional to means equal to this thing on the right times some constant that we don't care about. This is the constant that we don't care about. So this is read as the posterior distribution of theta, given the data, is proportional to the likelihood times the prior. That's the important part of base theorem. The posterior is proportional to the likelihood times the prior. So if we want to find our posterior distribution, we need to know our likelihood, and we need to know our prior. Those are the two things. And so if we recognize this product, the likelihood times the prior, if we recognize that as coming from a certain family of distributions, that's going to tell us what our posterior distribution is. So again, we can focus on these two terms here and a very common strategy is to identify that family or recognize that family of distributions that results from it. If it is a familiar family, sometimes it won't be. But if it is, we know what our posterior distribution is. So let's see an example of that. Let's say that the proportion theta of defective items in a large manufactured lot is unknown. So theta is our parameter of interest. And the prior distribution that we put on this parameter is uniform. uniform on the 0-1 interval. That's our prior for this proportion. Proportions are bounded between 0 and 1. And let's say we take a random sample now of n different items from this lot, and for each of those, the random variable X is going to be equal to 1 if that item is defective, and it's going to be equal to 0 otherwise. So what's the distribution of a data point? Bernoulli, with parameter. So each of these data points, each X sub I, is a Bernoulli random variable with probability theta.

So that's going to tell us our likelihood. Yep. Yeah, we have a prior on theta, but the likelihood is conditioned on the parameter. So the likelihood isn't treating it as random.

The randomness comes from the prior. Okay, so the individual data points are Bernoulli with probability theta. We have a uniform prior distribution on theta, and now we want to find the posterior distribution.

That's our goal in this problem. That down, goal, theta. I'm going to Put the parameter terms, terms for theta in blue, just to help us keep track of them. And also because I'm not good at bolding when I write, I'm going to underline to represent bold when I'm handwriting a problem. So if I type it out, it'll be bold, but if I'm handwriting, I underline that just represents a vector of bold. But that's our goal is the posterior distribution of the parameter theta given the observed data X. Okay? How do we get there? We have a prior, it's uniform, prior and what? Yeah, we're going to use base theorem that tells us how to get a posterior. That's our goal. And specifically, we're going to focus on what it's proportional to.

We're going to focus on the likelihood times the prior.

We know now on base theorem with this posterior distribution, pi of theta, and the data X is proportional to.

The likelihood, F of X, given theta, and is our prior, pi of theta.

We already said what the distribution of the data was to tell us our likelihood.

So we have f, bold vector x given theta. This is our likelihood or the joint distribution of x1 through xN given or conditioning on theta.

And because we have a random sample, we have independence, product i equals one and x sub i given the parameter theta.

And we know the probability mass function for an individual for newly random

variable. Right, we have p to the x times one minus p to the one minus x which returns p if x is one returns one minus p if x is zero so if we think about the sum of our x's right the number of defective units we have equal to theta to the sum of our x's right each case where x is one is going to return theta and then one minus theta and n minus the sum of our x is this is our likelihood and both of these terms involve theta so i'm writing lemon blue here.

Okay, that's the first ingredient to the posterior is the likelihood. The second is the prior.

Theta, rotation for the prior, right? And it's uniform. So this is just going to be equal to 1 times an indicator because we have constraint.

Times the indicator that theta is between 0 and 1.

So the prior density function is just 1 between 0 and 1 and it's 0 elsewhere, or 1 times this indicator function.

You could write it that way. It's the same thing. Yeah. Okay, so we have both the ingredients we need. We have our likelihood. We have our prior. Right. So we can get what our posterior is proportional to of theta given the data x is proportional now to theta the sum of the x's 1 minus theta or n minus the sum of x's the sum of non-defectives for the total exponent here is 1 times our indicator or ignore the 1 if you don't want it there so this is what our posterior distribution is proportional to so we know this distribution is bounded between 0 and 1 we get that from the indicator function So anything outside of 0 or 1 return 0. And then with between 0 and 1, this is what the probability density function looks like. If we think about a distribution only is supported between 0 and 1, it has a form. Click this. What distribution is that? Recognize that distribution. So binomial is for a discrete random variable. We have something continuous here, continuous between 0 and 1. Bernoulli is also discrete. Bernoulli is only equal to 0 or 1. Beta. Yeah. Keep in mind what we're looking at here. So we have the random variable to some power times one minus the random variable to some power. Let's go back to lecture one. Go to our table, beta one here. We have our random variable x to some power. We have one minus the random variable x to some power. We know the x, the random variable here, is bounded between zero and one. That's our restraint for the random variable. And then this is a constant. It's not a function of x. We don't really care about it. this piece, the terms that involve the random variable, they call that the kernel. So what we've done here is we've recognized the kernel. The kernel is only the part that has the red terms there. In our case, it's only the part that has the Theta's. The same thing, just here we were talking about data. We were using X as the random variable. In our case, we're talking about the parameter as the random variable, theta, but it's the same distribution.

Yeah, it could be any, you know, real number between zero and one.

We will talk about some other specific distributions, but for right now, these are the ones you should know. But we'll talk about other we'll talk about the T distribution and the kai square distribution that are useful for other things. But in terms of recognizing kernels, these are the ones that you really want. Yeah, even if you don't remember, you know, like the constants like this, it really helps to remember the kernels, because when you go to identify a posterior distribution, right, you write down the likelihood, you write down the prior, you look at the terms that involve the parameter, right, the easiest way to get your posterior. a distribution that you recognize is to recognize the kernel. So it does help a lot to at least get very familiar with the kernels here so that you can recognize them.

Yeah. Do you ever like a uniform posterior, you could probably construct like a weird case where it happens.

In practice, probably not.

So that's kind of the trick here. Recognize the kernel. We just recognized that this is the kernel of a beta distribution. So we know our posterior is going to be some beta distribution. We haven't identified what the parameters are, your question. So the marginal distribution is not a function of theta, right? So if theta changes from one value to another, this doesn't change, right? So the terms involving theta are this, and then it's just some constant multiplied in front of that. That constant doesn't change as we change a value of theta, though. So it's not important in terms of recognizing that the kernel of the distribution. Yeah, let me finish out this example real quick, and I think that'll kind of answer your question. So we've identified that this is going to be a beta distribution because we've identified the beta kernel. We'll write that down. This piece here recognize from the beta distribution. Going back real quick to that beta distribution. This one here, we have x to the parameter alpha minus 1 times 1 minus x to the parameter beta minus 1. So we need to rewrite things in terms of a parameter minus 1 in our example. I'm just going to rewrite this theta to the sum of the x's plus 1 minus 1, right? Because we had some of the x's, that's the same as sum of x plus 1 minus 1, 1 minus theta to the power minus the sum of the x's plus 1 minus 1 indicator function. So now we've actually written this in the form where we can identify our parameters, right? This was what we called alpha, this is what we called beta. It's the same form now as that beta distribution from our table. Why does this work? Well, we know now Theta given X, the posterior distribution. It is a distribution. Is a distribution. Meaning if we take this, this probability density function, and we integrate out the random variable theta, what should it integrate to? One. The probability distribution, so it integrates to one. So even though we didn't write down the constant here, we know what that constant has to be because it has to be whatever makes this integrate to one. The constant must be this.

It's a beta distribution. This is the constant that makes it integrate to one. That has to be the constant because we have a probability distribution.

It must be one over the beta function now. Alpha and beta goes in red and green.

Alpha has to be the normalizing constant because it has to integrate to one. So this is equal to 1, the beta function, sum of xI plus 1, the sum of xI plus 1. So we've identified the kernel, the kernel, and because we've identified the kernel, we know what the normalizing constant has to be to get it to integrate to 1. And that gives us everything we need to write down the full distribution or the full probability density function in this case. So pi theta x is going to be 1 over the beta function, which we remember uses the gamma functions. And so this is going to gamma n plus 2 or gamma sum of x plus one. Gamma minus the sum of x plus one. That's all the normalizing constant. And then we have our kernel theta of xI, 1 minus theta n minus sum of xI. Or you could add the plus one minus one if you want to see the parameter values explicitly. This is also add our indicator. This is our probability density function for the posterior distribution. Would the constant be different if you have the N choose X term? The process would have gone like this. So we would have written down something where there was the N choose X here. But that N choose X is a constant. It doesn't involve theta, right? So when we say the posterior is proportional to something, we ignore the constants. So it wouldn't get involved, it wouldn't get included in this line. No, it would be the same because this step would be the same. Because once we get here, we identify the kernel, we know it's a beta distribution, we find the appropriate normal. constant for that beta distribution. Yeah, yeah, yeah, I don't need you to evaluate what the gamma function is equal. You can write that. So that's just coming from, call from our first lecture. We talked about the gamma function, and then we talked about the beta function, right? It uses the gamma function, right? And we have one over the beta function, so we have gamma x plus y over gamma of x, gamma gamma of y. So it's just coming from this, and that beta function is the normally, one over that beta function is the normalizing constant for the beta distribution. So that's the really nice trick here, is because we know the posterior is proportional to the likelihood times the prior, we ignored everything that was constant with respect to theta, recognized a kernel, and that kind of gave us

the full posterior distribution. So the math was pretty easy. We just had to write it down and recognize something. And this is what we'll do quite often. Sometimes if you get a distribution that you don't recognize, then you might actually have to do that integration, find the normalizing constant yourself. that can be a pain, especially in really high dimensions. But all the cases we're going to look at here, we're going to focus on recognizing the constant. We're not really going to be integrating anything. But the flip side of that is you want to get very familiar with that table so that you can get really good at recognizing the kernels and identifying what the posterior distribution is going to be. So there's a very specific type of prior. So it's used all the time in practice, conjugate priors. And so for a given likelihood, If the prior distribution and the posterior distribution both belong to the same family, they're both beta, they're both normal, they're both gamma, whatever, if they both belong to the same family, we call that a conjugate prior distribution. So for these conjugate priors, it makes things really nice, really easy because we do our posterior updating, Bay's rule, get our posterior distribution, just by updating the parameters correspond to that distribution. I have an alpha and a beta for the prior. going to update the alpha and the beta in some like conjugate, right? So that we know the posterior is also going to have an alpha and beta. It's just going to be some different value using the data somehow. So these are conjugate priors, not the only types of priors that are used. Typically the most convenient prior, not always the best prior, but they're really nice in terms of this very easy updating. And so this is the type of prior we're going to focus on here. I'll go through a few examples. So the first is Bernoulli sampling. By that, I mean the distribution of the data is Bernoulli. And so this is called beta Bernoulli updating. So the beta is going to be the prior and the posterior. The Bernoulli is going to be the sampling model, the data model, the likelihood. Let's say that we have data x1 through xN. And it's a random sample from the Bernoulli distribution parameter p. We don't know p other than the restrictions that has to be between 0 and 1. And let's say the prior distribution of p is the beta distribution, the sum prior parameters alpha and beta.

This beta-burner-le-updating theorem says that then the posterior distribution of p, given the data, conditioning on the observed data points, will also be beta distributed, and it will have these parameters. So the first parameter, what was alpha, will be alpha plus the sum of the xes. We've updated alpha by adding the sum of the xes. And the second parameter, what was beta, will now be beta plus n, the sample size, minus the sum of the xes. Or we can think of this first one as alpha plus the number of successes. Second one has beta plus the number of failures. N minus the number of successes is the number of failures. Question. So let's go ahead and show this. We'll prove this. Okay, so we have beta Bernoulli updating. Bernoulli is the distribution of our data. So that's going to inform our likelihood. We know we need the likelihood and the prior to get the posterior. Okay, so F of X given p is our parameter now. This is our likelihood. We have a random sample, independent draws from the same distribution. All right, so this is equal to the product.

I equals 1 to n to the xI, 1 minus p to the 1 minus xI, or p to the sum of the x times one minus p to the n minus that sum.

Just like in the last example we did.

Okay, that's our likelihood. Prior distribution, pi, p, follows a beta It's just 1 over the beta function evaluated at alpha and beta. The prior parameters, p to the alpha minus 1, times 1 minus p to the beta minus 1, times an indicator function that p is between 0 and 1. This is our prior. We already wrote down the likelihood. Those are the two ingredients we need to get the posterior distribution. So now, posterior pi of the parameter p, given the data x, is proportional to the product of these two things. proportional to this is this.

Write down proportional to not equal to. It's proportional to p the sum of our x's times one minus p the power n minus the sum of x. It's one over this beta function. P alpha minus one. It's one minus p power beta minus one times our

indicator, p between zero and one. And that expression to one over the beta function, collect the terms involving p, collect the terms involving one minus p. So we're going to one. So we get p sum of x plus alpha minus 1, 1 minus p the power n minus the sum of x, beta is 1 times our indicator function. Posterior distribution is proportional to this expression. This first term in black is constant with respect to p. Don't care about it, ignore it, and get rid of it if we want to. These other three terms involve p. We need to consider them. This last one is just our indicator. It gives us our parameter set. values. And then again, we recognize this as the kernel of a beta distribution, just like in the last example.

So this is constant, concerned with it. The rest of this recognize a beta kernel. And so we know our posterior distribution is going to be a beta distribution.

We know that the parameter alpha from a beta distribution is everything before the minus one in the exponent for p, and the parameter for our Our second parameter of the beta distribution is everything in the exponent before minus 1 in the 1 minus p. So the posterior pi of p given the data x must be beta random variable. And I'm going to call these parameters alpha 1 and beta 1 to denote that we're updating. After we see the data, it's the posterior. And now parameters alpha 1 is equal to alpha our prior parameter plus the sum of x, beta 1 is equal to beta, our prior beta parameter plus n minus the sum of x.

So again, we just need to identify our likelihood, identify our prior. We know our posterior is proportional to the product of those two.

If we can recognize the kernel, that tells us what the distribution is. And you might notice that this is a very general version of our previous example. In the previous example here, we had a uniform prior, but we had Bernoulli data.

Well, the uniformed, distribution is a special case of the beta distribution where alpha is equal to one and beta is equal. This was the specific case where we had uniform, alpha one, beta one. This is a little more general now for any beta prior distribution, parameter alpha and beta generally now and Bernoulli. So you can use this result in homeworks, exams, whatever.

Just identify that you're using this beta-Bernulli updating theorem. Right. So just name that and you can use this result. Unless I specifically ask you to derive it.

So let's go. ahead and do an example now that uses this result. Suppose we have observed 10 independent flips of a coin, sample size 10. And of those 10 flips, we've observed eight heads. Our prior distribution for p, the probability of heads. So we're considering heads success, one, tails a failure zero.

The prior distribution for p is uniform or beta 1-1. And what's the posterior distribution for p?

And then also what's the posterior mean? What's the posterior variance? So this is just a direct application of this beta Bernoulli updating theorem. We can use this now. We've established this. 2 from above.

Use theorem 1. Okay, let's write down what we know from the problem. We're given the prior is uniform or beta 1-1. So I call alpha sub 0 1, beta sub 0 1. These are just the parameters from the prior distribution, the beta prior distribution.

We know our sample size n is 10, and we know sum of the data, the sum of X, is eight. We've had eight successful flips, eight landed on heads. And this is everything we need.

We know from the fact that we have beta-burnierly updating, the posterior is going to be beta with these parameters.

So that's it. Alpha-1, alpha-1, alpha-corresponding to our posterior distribution, which is also beta, is equal to alpha-zero plus the sum of our X's, or one plus eight, which is nine, beta-1 to beta-0, plus n minus the sum. of our x's or 1 plus 10 minus 8, which is 3. Using that theorem, we know posterior distribution pi, p given our data x follows a beta distribution, parameter alpha equal to 9, and beta equal to 3. That was the first part of the question. What's the posterior distribution for p? The second part of the question says, what's the posterior mean? When we know the distribution, the distribution tells us everything we want to know about a random variable. We can use the distribution to get things like the mean in the variance. So let's go ahead and do that. We want the posterior mean or the expectation of p given the data x, posterior mean. And we know for a beta distribution, or we could at least look up in our table if we didn't remember, that the mean of a beta random variable is alpha over alpha plus beta.

So this posterior mean is going to be alpha one, alpha one plus beta one. And we know these values.

This is 9 over 9 plus 3 or 0.75. Notice now the sample mean is 8 divided by 10 or 0.8. Posterior mean is 0.75, a little bit less than the sample mean, a little bit less than 0.8. And that should make intuitive sense because our prior distribution was uniform. Our prior distribution had mean 0.5. So we're landing somewhere in between the sample mean and the prior mean. That gives us our posterior mean. It falls closer to the sample mean than the prior mean, right, but it is between them. And then the last part of the question says, what's the posterior variance?

And again, we can look this up in our table if we don't have it memorized. Alpha one, beta one, alpha one, beta one, quantity squared.

Alpha one plus beta one plus one plus one. We know the values of these is nine times three, 12 squared, 13, which simplifies to 27 and 72. or about 0.0144. So the fact that we had a conjugate prior from this beta Bernoulli updating made this problem very easy. We had to identify that we had Bernoulli data, the beta prior distribution, and then our theorem tells us we know the posterior is going to be beta as well, right?

And we just update those parameters. We update alpha by adding the sum of the data. We update beta adding N minus the sum of data or the number of failures. And once we have our posterior distribution, we can do anything that we do with a distribution, means, variances, and so on. Questions here? Okay, let's look at another case of a conjugate prior. We have poisson distributed data, count data. This theorem is called gamma poison updating. Suppose we have a random sample of data, x1 through xN, coming from a poisson distribution, and that poison distribution has parameter lambda. We don't know lambda. It does have the constraint that has to be positive. And then we put a prior distribution on lambda. If that prior distribution is gamma, call that the gamma distribution has parameters alpha and beta. If this prior distribution for lambda is a gamma distribution, the posterior distribution of our parameter lambda given the data will also be a gamma distribution. Conjugate. Gamma prior leads to gamma posterior when we have poisson data, poisson likelihood. And in this case, the parameters will be alpha plus the sum of the data, updating alpha by adding the sum of the data and beta plus n update beta by adding the sample size and we can show this just like we did in the beta Bernoulli case we need to start with our likelihood f of x given lambda a random sample independent so it's going to be a product and the data is poisson f is poisson for an individual data point so we need the poisson probability mass function here not memorized yet we could look up in our table i'll just give it to you here.

So this is going to be lambda to the power xI times e to the negative lambda divided by xI factorial.

Both terms of the numerator involve the parameter lambda. The denominator does

not. And we could simplify this product to be lambda to the power sum of xI, e to the negative n is lambda times the product of these one over x factorial terms.

This is our likelihood when we have Poisson data.

We're putting a gamma prior distribution on our parameter lambda. So pi here, it would be a gamma PDF.

Probability density function is beta, the alpha over gamma alpha, lambda, the alpha minus 1 times E the negative lambda is beta indicator function that our parameter lambda is positive.

Gamma probability density function for the prior distribution on lambda. Okay, we have everything we need now. I try to identify the posterior distribution using base theorem. Base theorem tells us pi, lambda, given the data, x proportional to the likelihood times the prior, which we've already identified. So proportional to sum of x to e the negative n times lambda over xI, our likelihood times the prior to the alpha minus 1, E negative lambda, beta, beta to the alpha, or gamma, alpha, and our indicator function base theorem. All of the terms involving the parameter lambda are given in blue. Terms that don't involve it are constant, given in black. We don't care about those. The functionality is equal to product. I equals 1 to n, 1 over xI factorial, beta to the alpha or gamma of alpha. Stuff we don't care about the constant. Then collecting terms that involve lambda and collecting terms that involve E to the negative lambda, we get lambda alpha plus sum of xI minus 1 times E the negative beta n times lambda times our indicator function. So we don't care about the constant. We can recognize this as the kernel of a gamma distribution. This is the parameter alpha of that distribution. This is the parameter beta of that distribution. the second parameter.

It tells us everything we need to know. So pi, lambda, given the data X, the posterior distribution, must be gamma.

There's alpha one, beta one, alpha one, alpha, alpha, the prior parameter, the sum of our data X, beta one, the second parameter of this gamma distribution is equal to beta, the prior parameter, plus our sample size n.

So again, we've shown this now. We have this theorem. We can use it when we like. So whenever you identify, that you have a random sample of Poisson data and you have a gamma prior distribution for that parameter. We can use our gamma Poison updating result and we know that the posterior distribution is also going to be gamma with these updated parameters.

Let's do a quick example using that result now.

So a store owner models customer arrivals as Poison with unknown rate Lambda per hour. She assigns Lambda, a gamma prior distribution. The parameter is alpha 3 and beta 2. And let's let this data point X be the number of customers that arrive in a specific one hour period. We only have one observation here, one data point. X equals 3 is observed. Find the posterior distribution of lambda, a rate of people per hour, lambda. And then what's the prior variance? The posterior variance. So again, we're just going to use that theorem. We have that theorem now. Let's make use of it. theorem 1.3 above. Let's write down what we know. We know alpha 0 from our prior distribution. 3. We know beta 0 coming from our prior distribution is 2. We know our sample size, N is 1. We have one observation, one, one hour window with customers. And we know the sum of X is just the single data point X is 3. But everything we need now to find the posterior distribution. using our gamma-poison updating result. So alpha-1 corresponding to the posterior distribution is alpha-0, 3, the sum of x. 3.3, 6, beta-1, beta-0, 2, plus our sample size n 2 plus 1 is 3. And that's it.

We have our posterior. Posterior distribution of lambda, given our single data point X, is also gamma distributed, but its parameters are 6 and 3, whereas the

priors were 3. That's the posterior distribution of lambda. We're asked then, what is the prior variance? We know the prior distribution is gamma distributed, parameters alpha and beta. We could look up the variance for a gamma distribution from our table. Just give it to you here, the variance alpha over beta squared. Again, just coming from that table, three-fourths. The zero is three, beta-zero is two, the beta-zero squared is four. That's the variance of our prior distribution. tells us some measure of spread or uncertainty in our prior. Now let's do the same for our posterior distribution. The variance of the posterior or the variance of this random variable lambda given the data point, conditioning on the data point, is just the variance of a gamma distribution with parameters six and three now. Alpha 1 over beta 1 squared, 6.9, 2 thirds. But our posterior variance is a little bit smaller at prior variance. It should make intuitive sense. We've learned something about, the parameter through the collection of data. We should be reducing our uncertainty. In this case, we only collected one data point, so we didn't reduce it that much, but we are reducing it. We're collecting data.

We're learning something about the parameter here. So again, another case where conjugacy makes it really easy to find posterior distribution, very convenient to do so. The last example I want to go over is normal sampling. So suppose that we have a random sample X1 through XN coming from a normal distribution now.

And let's say the mean of that normal distribution, mu, is unknown. But let's say the variance is known. So we're only considering one parameter here, one unknown parameter here. So the mean's unknown, but the variance is known. So we need to put a prior distribution on that unknown mean. And let's say we put a normal prior distribution on it. So that normal prior distribution has its own parameters. It's a normal distribution. So it has a mean parameter and a variance parameter. So we'll call that mean parameter of the prior mu mu sub-0, we'll call the variance of that normal prior, sigma squared sub-zero. So in this case where we have normal data, unknown mean, known variance, and we put a normal prior on that unknown mean, then the posterior distribution of mu given the data will also be normal, and it will have updated parameters, mu-1, and sigma-squared 1, calculated as seen here.

So again, an updating rule, this one's a little more complicated, but, right, you can just plug in the values. Right, We know sigma squared. It's part of the assumption here. We choose the prior mean, mu sub zero, N. We're going to know how many data points we observed, and X bar is the sample mean. So we know all these values, or we will know them after we collect data anyway and specify our numbers. I'm not going to go through the derivation for this one. It's a little bit more involved. The book does go through it if you'd like to see it, the textbook. Basically it involves, like we did before, you know, getting rid of any constants, collecting the terms, but then you have to do a completing the square step. Go ahead and look at that if you want. But again, you can use this theorem if you identify, right, that the conditions are met, you have normal data, unknown mean meo, known variance, and you put a normal prior on that unknown mean mew, you can use these updating rules. So let's do an example of this one. So they've sampled 20 nationally prepared foods and compared the stated calorie contents program for those foods from the labels to the actual calorie content determined in the laboratory. Let's say we model the conditional distribution of the differences. So we're modeling the data here. Our likelihood, the difference is given mu as normal. Mean mu is unknown, but we know the variance is 100 here. So we have our normal sampling model. We have a known variance, unknown mean. And then we're going to put a prior distribution on mu. It's also normally distributed. The prior mean is zero. Prior variance is 60. And we collect 20 of these differences. And we find that their average, their sample, 125.

average is 0. That's the posterior distribution. Again, we've met the conditions for this normal updating from our theorem.

So we can go ahead and use that theorem. Use theorem in 0.4 above. We know D, normal, mean mu, invariance of 100.

xI given mu, i. This is the distribution of our data, condition on the parameters or our likelihood.

And we know parameter mu has a normal prior distribution with mean zero and variance 60. Let's just identify all of our pieces. We need the sample size, N, that's equal to 20. We need sigma squared. The variance associated with the data model, the sampling model, which we're told is 100, mu zero, the mean of our prior distribution, which is zero. We need sigma squared zero, the variance of our prior distribution, which is 60, and we need X bar, our sample mean, which we're told is 0.1.25. These are all of the ingredients that go into our updating rule for normal sampling. So we're going to use that theorem now. Calculate mu 1, the mean of our posterior distribution, 100, sigma squared, 0, mu sub 0, plus our sample size n20 times our prior variance of 60 times our sample mean 0.125. That's our numerator for mu 1, and the denominator is sigma squared 100 plus our sample size n20 times our prior variance. variance 60.

And that all comes out to 0.114. Other piece, the other parameter that we need, sigma squared sub 1, the posterior variance, and that's equal to 100 sigma squared times our prior variance 60 or sigma squared 100 plus our sample size 20 times our prior variance 60, which comes out to 4.

And that's it. The normal distribution has two parameters, the mean on the variance, and we've identified them. Pi with mu, given the data X, is normal with mean 0.11, 4, variance 4.62. A case where a conjugate prior, a normal prior, on a normal mean, leads to very easy updating rules. So I want to just finish by talking about the effects of the prior. So I want to do this specifically for this last case, this normal sampling case. So what I'm doing here is I'm sampling from a normal distribution, where the setup is sampling from a normal distribution, where we don't know the mean, right, we want to estimate the mean mu, but we know the variance is equal to one, it's equal to one, unknown mean mu, and we're going to sample 10 data points. Our sample size is 10, right? And let's say when we sample those 10 data points, the sample mean is 3. And let's think about, or we're going to see what happens when we choose different priors and how those priors affect our posterior, how they change our posterior, right? The data is the same in every case, right? In every case, we've collected 10 data points and the sample mean is 3, but what happens to the posterior for each different prior that we've chose? All three priors are centered at zero. They have the same prior mean, which is zero, but their prior variances are different. We're going to consider one half, one, and five. Three different priors with three different prior variances. And this is what happens. These blue curves are the prior probability density function. You see, this top one is the most concentrated, this bottom one is the most spread out. This top one has the smallest prior variance. This bottom one has the largest prior variant. The red is the likelihood, specifically the scaled likelihood. Question earlier, could we scaled the likelihood. I scaled it here to make it visually representative compared to higher in the posterior. But we see it the same in every case. This red curve doesn't change. The data that we observed was the same in every case. And then finally, the focus here is this green piece. That's the posterior distribution. How does the posterior distribution change for each of these three different? priors. So if you look at the bottom, when we had the really large prior variance, really spread out prior, the posterior matches very, very closely to the scaled likelihood. Basically, it's just the data that's driving this. The prior is not doing a whole lot. But as you go up to this small variance prior distribution, this more concentrated prior distribution, right? We see that this posterior distribution is pulled to the left, shrunk to the left, towards the prior distribution. In all three cases, we see the posterior is somewhere. between the likelihood and the prior, where in between depends on the prior that we chose, right? It gets much closer to the likelihood. If we have a very large prior variance, it gets much, not much, but it gets closer to the prior distribution when we had a smaller prior variant. So that's kind of the impact of the prior distribution. Here we're saying we're pretty confident about our prior distribution, right? So we're going to give more weight to the prior in that case. Here we're saying we're not really

confident. We have a lot of uncertainty in our prior, we're giving much more weight to the data. Just some effects of the prior I want to talk about here. When the sample size N is small, the posterior mean is going to be much closer to the prior mean. Pryor is going to get more weight when we have little data. The flip side of that is when we have a lot of data when the sample size is large, the posterior mean is going to get pulled much closer to the sample mean. The more data we have, the more weight we're giving to the data, we've learned about the data, the less we rely on our prior. The prior variance, when this is small, like in this top row here, we give more influence to the prior in terms of our posterior. Posterior is going to be pulled closer to the prior mean. If the prior variance is large, it's the opposite. It's pulled closer to the likely. And then lastly, as the sample size increases, the posterior variance is going to decrease. We kind of saw that in one of the examples. As we collected that Poisson data point, the posterior variance was smaller than the prior variance. That's another consequence of increasing the sample size. Any questions here? Yeah. Yeah. So if we look at our normal updating rule here, the posterior variance is a function of the prior variance. It does depend on sigma squared there. In this case, it's very slight. The bigger thing to notice here is the movement from left to right, the centering, but you're right, there is a little bit of impact our variance. One more thing I'll mention real quick. We said, M1 from our updating rule, is sigma squared times zero. mu 0 plus n sigma squared 0 x bar over sigma squared plus n sigma squared sub 0. We could rewrite this sigma squared over sigma squared plus n sigma squared 0 is mu 0 plus n sigma squared sub 0 over sigma squared sigma squared sub 0 times x bar. Rearranging that equation. And now we have something that's pretty interpretable. This is saying the posterior mean is a weighted average of the prior mean and the sample. mean. These weights sum to one, right? But it's a weighted average of these two things. And what did we say? We said when the sample size n is small, the posterior mean will be closer to the prior mean. If the sample size n is small, this numerator is going to be small, this weight's going to be small. So we give less weight to the data, more weight to the prior. And the reverse of that is true. If n is large, this weight becomes large, we give more weight to the data, x bar, the sample mean. So we can kind of see mathematically that these first two things are True.

The prior variance, sigma squared, is very, very small. Here, when this is very, very small, this will be close to sigma squared over sigma squared, close to one, right? So we'll be giving a lot of weight to the prior mean if our prior variance is very small, and vice versa. If our prior variance is very large, this weight will get smaller. We give less weight. So just another kind of way to see these interpretations mathematically by rewriting this as a weight of average, the prior mean, and the sample mean. I'll end there. I'll stick around for a bit if there's questions. Otherwise, Have a nice weekend and I will see you on Tuesday.